

DATABASE MANAGEMENT SYSTEMS (IT 2040)

Question 1

What is a Database Management System (DBMS)?

- A) A collection of related data
 - B) A general-purpose software system that facilitates defining, constructing, manipulating, and sharing databases
 - C) A file processing system for storing data
 - D) A programming language for creating databases **Answer: B**
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Question 2

Which of the following is NOT an advantage of the database approach over file processing systems?

- A) Data independence
- B) Enforcing integrity constraints
- C) Simpler implementation for small-scale applications
- D) Concurrent access by multiple users

Answer: C

Question 3

In the database design process, what is the FIRST phase?

- A) Conceptual database design
- B) Logical database design
- C) Requirement collection and analysis
- D) Physical database design

Answer: C

Question 4

What type of entity cannot be uniquely identified alone in a domain?

- A) Strong entity
- B) Weak entity

□

- C) Composite entity
- D) Abstract entity

Answer: B

Question 5

In an ISA relationship hierarchy, what does "disjoint" mean?

- A) All entities in subclasses are also in the superclass
 - B) An entity cannot belong to more than one subclass simultaneously
 - C) An entity must belong to at least one subclass
 - D) Subclasses share all attributes with the superclass **Answer: B**
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Question 6

What is a fan trap in ER modeling?

- A) A missing relationship between entity types
- B) An ambiguous pathway between certain entity instances due to multiple one-to-many relationships fanning out from the same entity
- C) A relationship that connects more than three entities
- D) An entity with too many attributes

Answer: B

Question 7

Which constraint determines whether entities in subclasses collectively include all entities in the superclass?

- A) Overlapping constraint
- B) Participation constraint
- C) Covering constraint
- D) Cardinality constraint

□

Answer: C

Question 8

What is the identifying relationship set requirement for weak entities?

- A) Many-to-many relationship with the owner entity
- B) One-to-one relationship with the owner entity
- C) One-to-many relationship where the weak entity has total participation
- D) Optional participation in any relationship

Answer: C

Question 9

When should a ternary relationship be used instead of binary relationships?

- A) When you need to connect exactly three entities
- B) When there is a common attribute that needs all three entities together
- C) When binary relationships are too complex
- D) When you want to simplify the ER diagram

Answer: B

Question 10

What does aggregation in EER modeling allow you to do?

- A) Combine multiple entities into one
- B) Indicate that a relationship set participates in another relationship set
- C) Create subclasses from a superclass
- D) Define composite attributes

Answer: B

Question 11

A chasm trap occurs when:

- A) Too many relationships fan out from one entity
- B) A model suggests a relationship should exist but it doesn't actually exist
- C) An entity has no primary key
- D) Multiple entities share the same attributes

□

Answer: B

Question 12

In the conceptual database design phase, which model is commonly used?

- A) Relational model
- B) Object-oriented model
- C) Entity-Relationship (ER) model
- D) Hierarchical model

Answer: C

Question 13

What does "total specialization" mean in an ISA hierarchy?

- A) All subclasses overlap with each other
- B) Every entity in the superclass must belong to at least one subclass (covering constraint exists)
- C) Subclasses cannot share entities
- D) All attributes must be specialized

Answer: B

Question 14

Which of the following is an aspect to consider during requirement analysis?

- A) Database normalization rules
- B) Hardware specifications
- C) What data is to be stored in the database
- D) Network topology

Answer: C

Question 15

What is the main difference between specialization and generalization?

- A) Specialization is top-down (defining subclasses), generalization is bottom-up (identifying commonalities)
- B) Specialization uses ternary relationships, generalization uses binary

□

□

C) Specialization is for weak entities, generalization is for strong entities

- D) There is no difference, they are the same concept **Answer: A**

SECTION B: MULTIPLE ANSWER QUESTIONS (Select ALL that apply) Question 16

Which of the following are advantages of the database approach? **(Select all that apply)**

- A) Data independence
- B) Efficient data access
- C) Lower initial development cost than file systems
- D) Enforcing integrity constraints
- E) Restricting unauthorized access
- F) Providing backup and recovery

Answers: A, B, D, E, F

Question 17

During requirement collection and analysis, what should database developers do? **(Select all that apply)**

- A) Interview clients
- B) Analyze documents and reports
- C) Design the physical database schema
- D) Identify unclear or incomplete requirements
- E) Eliminate redundant information

Answers: A, B, D, E

Question 18

Which of the following are phases in the database design process? **(Select all that apply)**

- A) Requirement collection and analysis
- B) Software testing
- C) Conceptual database design
- D) Schema refinement
- E) Network configuration
- F) Physical database design
- G) Security design

Answers: A, C, D, F, G

Question 19

For a weak entity to be valid, which restrictions must hold? **(Select all that apply)**

- A) It must participate in a one-to-many relationship with an owner entity
- B) It can be uniquely identified on its own
- C) It must have total participation in the identifying relationship
- D) Each weak entity can have multiple owner entities
- E) The identifying relationship is one-to-many with the weak entity on the many side

Answers: A, C, E

Question 20

What are characteristics of the database system approach compared to file processing? **(Select all that apply)**

- A) Self-describing nature
- B) Insulation between programs and data
- C) Support for multiple views of data
- D) Single-user access only
- E) Sharing of data and multiuser transaction processing
- F) No security features

Answers: A, B, C, E

Question 21

In EER modeling, ISA relationships can have which constraints? **(Select all that apply)**

- A) Overlapping constraint
- B) Foreign key constraint
- C) Covering constraint
- D) Disjoint constraint
- E) Cardinality constraint

Answers: A, C, D

Question 22

Which statements about ternary relationships are correct? **(Select all that apply)**

- A) They involve exactly three entities
- B) They can represent common attributes that need all three entities together
- C) They should replace all binary relationships
- D) They can coexist with binary relationships if both have different meanings ☐ E) They are the same as aggregation

Answers: A, B, D

Question 23

What can cause design traps in ER modeling? **(Select all that apply)**

- A) Fan traps with ambiguous pathways
- B) Proper normalization
- C) Chasm traps with missing relationships
- D) Multiple one-to-many relationships fanning from the same entity
- E) Optional relationships creating ambiguous paths

Answers: A, C, D, E

Question 24

Which of the following are true about aggregation in EER models? **(Select all that apply)**

- A) It allows a relationship set to participate in another relationship
- B) It contains two independent relationships
- C) It is the same as a ternary relationship
- D) It can be used to model complex dependencies
- E) It represents performance-based relationships like awards for player participation

Answers: A, B, D, E

Question 25

What aspects should be considered during requirement analysis? **(Select all that apply)**

- A) What data is to be stored

- B) What programming language to use
- C) What applications are to be built
- D) What operations have to be performed
- E) Which hardware vendor to choose

Answers: A, C, D

Question 26

Which of the following are direct or indirect advantages of using a DBMS? (Select one or more)

- A) Maintaining integrity of data
- B) Restricting unauthorized access to data
- C) Reduced program maintenance
- D) Providing fast access to data
- E) All of the above

Answer: E

Question 27

Which of the following are direct or indirect advantages of using a DBMS? (Select one or more)

- A) Maintaining integrity of data
- B) Restricting unauthorized access to data
- C) Reduced program maintenance
- D) Providing fast access to data
- E) All of the above

Answer: E

Question 28

Which of the following are direct or indirect advantages of using a DBMS? (Select one or more)

- A) Maintaining integrity of data
- B) Restricting unauthorized access to data
- C) Reduced program maintenance
- D) Providing fast access to data
- E) All of the above

Answer: E

Question 29

Which of the following is/are TRUE about databases and database management systems? (Select one or more)

- A) Databases are used to only store complex information
- B) All databases are computerized
- C) Defining a database involves specifying the data types, structures, and constraints on data
- D) DBMS is a special purpose software that is capable of structuring, storing, and programming data
- E) Database definitions are stored in a database catalog

Answer: C, E

Question 30

Which of the following is the correct order of processes involved in developing a database?

- A) Requirement collection and analysis → Conceptual database design → Schema refinement → Logical database design → Security design and Physical database design
- B) Requirement collection and analysis → Conceptual database design → Schema refinement → Logical database design → Physical database design and Security design
- C) Requirement collection and analysis → Physical database design → Conceptual database design → Logical database design → Schema refinement → Security design
- D) Requirement collection and analysis → Conceptual database design → Logical database design → Schema refinement → Physical database design and Security design
- E) Requirement collection and analysis → Conceptual database design → Logical database design → Schema refinement → Security design and Physical database design

Answer: D

Question 31

Which of the following are not examples of program–data independence (insulation between program and data)? (Select one or more)

- A) Being able to hide from users where the actual data are stored.
- B) Being able to access data using programs written in different programming languages.
- C) Being able to define which users are able to access data.
- D) Being able to improve the performance of the database without affecting the data.
- E) Being able to add columns to a table without affecting user queries.

Answer: C

Question 32

Which of the following is/are intensions (intentions) of a database developer during the requirement collection and analysis phase? (Select one or more)

- A) Finding the names of the people who will be developing the applications to access the database.
- B) Finding relationships among data in the organization.
- C) Finding data to be stored in the organization.
- D) Identify the number of concurrent users who will be using the database.
- E) Identify different types of data retrievals to be performed on the database.

Answer: B, C, D, E

Question 33

Which of the following are direct/indirect advantages of using a DBMS? (Select one or more)

- A) Maintaining integrity of data
- B) Restricting unauthorized access to data
- C) Reduced program maintenance
- D) Providing fast access to data
- E) All of the above

Answer: E

Question 34

Which of the following are NOT examples of program-data independence (insulation between program and data)? (Select one or more)

- A) Being able to add columns to a table without affecting user queries
- B) Being able to access data using programs written in different programming languages
- C) Being able to improve the performance of the database without affecting the data
- D) Being able to define which users are able to access data
- E) Being able to hide from users where the actual data are stored

Answer: D

Question 35

Which is/are NOT a characteristic(s) of the database approach? (Select one or more)

- A) Support of a single view of the data
- B) Self-describing nature of a database system
- C) Data collection
- D) Insulation between programs and data

- E) Increasing redundancy of data

Answer: A, C, E

Question 36

University rooms scenario Which of the following are true?

- A) A group cannot exist without a batch
- B) There are no descriptive attributes for relationships in the diagram
- C) There are two identifying relationships in the diagram
- D) Allocated can be considered as a ternary relationship
- E) The entity batch has a composite key

Answer: A, C, D, E

Question 37

Which of the following are direct or indirect advantages of using a DBMS? (Select one or more)

- A) Maintaining integrity of data
- B) Restricting unauthorized access to data
- C) Reduced program maintenance
- D) Providing fast access to data
- E) All of the above

Answer: E
